

CHARTS AS EPISTEMIC ACTIONS: EXPERIENCE-DRIVEN CHART USE IN AGENTIC DATA ANALYSIS

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ABSTRACT

Charts are important cognitive tools that help human analysts identify trends, patterns, and anomalies. Although data agents can generate and interpret charts, the analytical utility of charts in agentic data analysis remains underexplored. In this work, we first conduct a probe study based on annotated questions to examine agents’ ability to interpret chart evidence and their chart-use behavior in open-ended analysis. We find that the evaluated agents effectively extract useful evidence from charts and tables, but differ substantially in whether they actively use charts during analysis. Moreover, charts and text exhibit task-dependent complementarity. These findings suggest that charts should not serve merely as post-hoc presentation tools, but should form an intrinsic part of data agents’ reasoning. Building on this insight, we propose an experience-driven method for chart use in agentic data analysis. The method distills analysis and representation experiences from conditioned trajectory rollouts using outcome scores and step-level attributions. The validated experience package helps agents learn how to construct, select, and use chart evidence while mitigating representation-use biases. Experiments on three held-out benchmarks show that our method encourages agents to actively use charts for analysis and significantly outperforms baseline methods.

1 INTRODUCTION

Data agents transform analytical goals into sequences of data-grounded actions and insights. Advances in ReAct-style reasoning-acting loops and tool use (Yao et al., 2023) enable these agents to plan, execute, inspect, and revise increasingly complex analytical workflows.

Despite this progress, most ReAct-style data agents remain text-centric. SQL queries and Python programs specify analytical operations, but their results are typically converted into textual observations before being returned to the agent. Human analysts work differently, moving repeatedly between numerical summaries, tables, and charts to expose different evidence and guide subsequent operations. This contrast creates a fundamental gap between current data agents and human analysis. It raises a central question for agentic data analysis, whether an agent should rely on text, use charts as human analysts do, or choose between them as the analysis unfolds.

Existing methods address only parts of this decision. Most data agents reason over textualized execution results and use charts primarily as final report artifacts. Chart-generation methods optimize visualizations requested by a user or an agent (Dibia, 2023; Yang et al., 2024; 2025; Namgoong et al., 2025; Lu et al., 2026; Wang et al., 2025), whereas chart-understanding methods assume that a chart is already available and focus on extracting information from it (Masry et al., 2023; Han et al., 2023; Xu et al., 2023; Liu et al., 2023a;b; Zhang et al., 2024). Recent studies introduce visual representations into reasoning or select among predefined formats (Kaur et al., 2026; Das et al., 2025; Zhou et al., 2025; Ho et al., 2026; Xing et al., 2026; Lin et al., 2026), but chart use remains prescribed, dependent on agent-specific preferences, or disconnected from the current analysis state. Consequently, existing methods do not provide a reliable mechanism for choosing between text and chart evidence during open-ended analysis.

An effective data agent should select both the analytical operation and the evidence representation that best fit the question and current analysis state. Before designing such an agent, we must establish whether charts provide genuine analytical value, how their utility differs from text, and whether current agents exhibit a gap between understanding charts and using them effectively. We therefore

construct PROBETASK to evaluate agents under controlled text and chart interfaces across analytical tasks and cognitive demands. Two main findings emerge. First, agents can extract sufficient evidence from charts, yet access to plotting tools does not produce reliable chart use, revealing a pronounced *capability–utility* gap. Second, charts and text serve as complementary evidence representations, with charts supporting trends, patterns, and anomalies, and text supporting precise reading, conditional verification, and evidence integration.

Motivated by these findings, we propose an experience-driven method for chart use in agentic data analysis. For each PROBETASK, the method generates conditioned trajectory rollouts and uses outcome scores and step-level attributions to distill an experience package. The package contains analysis experience for selecting and verifying analytical operations, and representation experience for deciding when and how intermediate evidence should use text, charts, or both. Candidate packages are retained only when they improve validation performance. During downstream analysis, the frozen experience package guides the agent in selecting analytical operations and evidence representations without fixed representation-selection rules.

In summary, our main contributions are as follows:

- We systematically probe the analytical utility of charts for data agents on synthetic trajectories (PROBETASK), revealing a gap between chart understanding and productive use of open plotting, and summarizing task-dependent complementarity between charts and text as evidence representations.
- We propose an experience-driven method that distills analysis and representation experiences from conditioned trajectory rollouts with outcome scores and step-level attributions, enabling agents to learn both what evidence to construct and how to represent and use it without fixed representation-selection rules.
- Experiments on three external data-analysis benchmarks (DAComp-DA, DDR-Bench, and DS Bench) with six LLM backbones demonstrate that our method consistently outperforms matched baseline agents; ablations on single experience types, chart disabling, and a fixed experience package quantify the contribution of experience distillation and visual representation.

2 CHARTS IN THE ANALYTICAL LOOP

We use PROBETASK to determine whether charts provide genuine analytical value to data agents and how their utility differs from text. We first describe the probe design and evaluation protocol, then examine chart-based analysis, open-ended representation choice, and the complementary strengths of text and charts.

2.1 PROBE DESIGN

Task construction. To isolate how intermediate representations affect agent reasoning, we synthesize PROBETASK from InsightBench (Sahu et al., 2025), which provides enterprise tables, open-ended questions, executable reference analyses, and target insights. From the first 50 source-data flags of InsightBench, we synthesize 200 PROBETASK tasks by treating each valid insight as an independent task. Each task contains a raw CSV table, dataset description, analytical question, reference analysis, and target insight. The agent receives the table, question, description, schema, and first three rows, while the target insight remains hidden for evaluation; it must select and execute the required operations and return a direct natural-language answer.

Annotations and evaluation. We annotate each task along two complementary dimensions for post-hoc analysis. *Question type* describes the requested answer, including numeric lookup, composition, rank or extreme, group comparison, temporal change, relationship, anomaly or risk, diagnosis, and prediction; *cognitive load* describes the primary operation, including locate, scan and select, compare, recognize a pattern, interpret a relation, integrate dimensions, and conditional reasoning. An LLM assigns the initial labels and a human annotator reviews them; these labels support slice analysis only and never define representation-selection rules. A common GPT-5.6 Sol judge scores each answer against the question and target insight on a 0–4 scale, and we report the mean normalized score on

Table 1: Interface ablation on PROBETASK with GPT-5.6 Sol.

Interface	Score
No-tool	29.4
Text-only	77.2
Chart-only	78.4
Free-style	77.6

Table 2: Free-style results across six LLM backbones.

LLM backbone	Score	Representation use (# tasks)			ReAct cost	
		Text	Chart	Both	Rounds	Tokens
Claude Opus 5	83.1	43	0	157	3.9	17,401
Claude Sonnet 5	79.2	102	0	98	4.0	18,724
GPT-5.6 Luna	76.2	198	0	2	3.1	6,835
GPT-5.6 Sol	77.6	198	0	2	2.9	6,392
DeepSeek V4.1 Flash	80.2	143	0	57	6.2	30,051
HY3	73.1	198	0	2	3.8	7,725

[0, 100]. Slice analyses use task-paired differences with flag-clustered bootstrap intervals (Efron & Tibshirani, 1993) and randomization tests.

Agent interfaces. All settings share the same vanilla ReAct agent with one general-purpose Bash tool. Each call starts a fresh process, while files persist in a shared working directory and the CSV is exposed read-only. We hold the task, model, initial context, interaction limit, and tool budget fixed across groups; only the analytical interface differs. **No-tool** removes execution tools, **Text-only** exposes stdout and stderr but prohibits plotting, **Chart-only** exposes generated images and stderr while hiding stdout, and **Free-style** returns stdout, stderr, and images and permits both text and charts.

On the full suite ($n=200$), we run an interface ablation with GPT-5.6 Sol across all four settings and compare six model backbones under Free-style; Tables 1 and 2 report the corresponding results. The same GPT-5.6 Sol judge scores all runs.

2.2 EMPIRICAL FINDINGS

Agents can conduct analysis through charts. Table 1 reports mean scores for the four agent settings. GPT-5.6 Sol scores 29.4 without tools, 77.2 with text, and 78.4 with charts. The chart condition performs comparably to text and improves over no-tool by 49.0 points, showing that agents can extract sufficient evidence from charts to answer analytical questions.

Open-ended representation choice exhibits strong model-specific bias. Table 2 compares representation use and interaction cost under the Free-style interface. Claude Opus 5 combines text and charts on 157 tasks, Claude Sonnet 5 on 98, and DeepSeek V4.1 Flash on 57, whereas GPT-5.6 Luna, GPT-5.6 Sol, and HY3 do so on only 2 tasks each. Chart-use frequency also has no monotonic relationship with performance, as models with similar use frequencies attain different scores. Thus, the same text-and-chart interface does not induce a common representation-selection strategy.

Charts and text exhibit task-dependent complementarity. Figure 1 breaks down the four settings by question type and cognitive load. Charts score 0.799 on temporal-change questions, exceeding text by 0.125, and lead by 0.096 on anomaly or risk and by 0.042 on relationship analysis. They also lead by 0.064 on pattern recognition, by 0.029 on relational explanation, and by 0.075 on multidimensional integration. The two representations perform similarly on composition or distribution, ranking or extrema, and scan and selection. Free-style performs below the chart condition on temporal change and pattern recognition, showing that access to both representations does not automatically preserve the benefit of chart evidence.

Text scores 0.472 on numerical queries compared with 0.250 for charts, and 0.604 on lookup tasks compared with 0.521 for charts. It also leads by 0.032 on group comparison and by 0.018 on comparative judgment. On conditional reasoning, text scores 1.000 while charts score 0.750, although this slice contains only four tasks. Text therefore has a relative advantage in exact reading, information location, and conditional verification.

Together, these results reveal a gap between chart-understanding capability and effective chart use. Agents can extract evidence from self-generated charts, but plotting access alone does not ensure reliable use or coordination with text. The appropriate representation depends on the task and current analysis state, motivating the experience-driven method developed next.

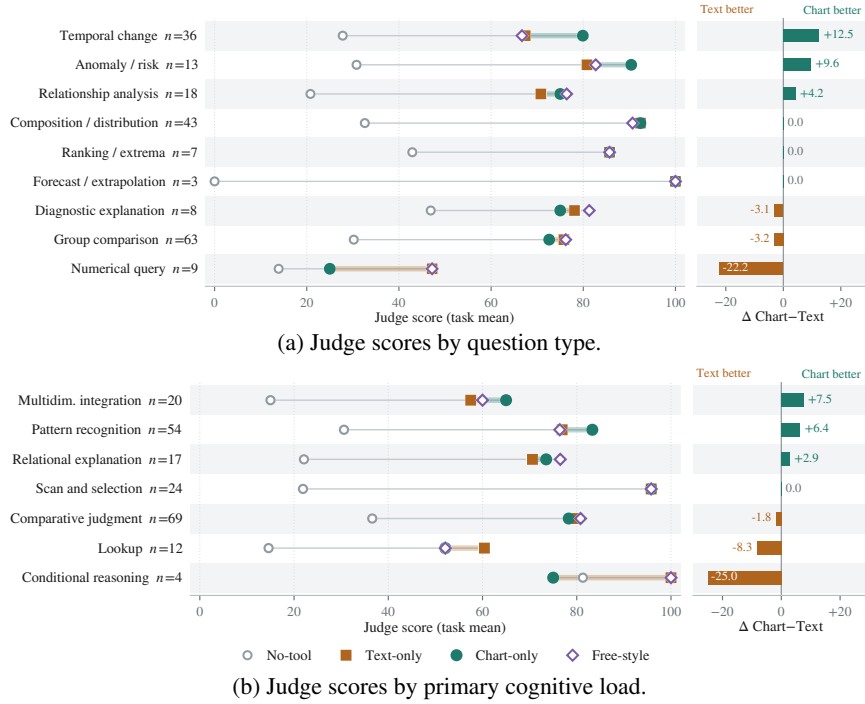


Figure 1: Slice analysis of the four agent settings on PROBETASK ($n=200$). Markers show slice means; the right column gives Chart-Text Δ (faded when $n < 10$). The bottom row of a aggregates all tasks.

3 LEARNING TO USE CHARTS IN AGENTIC DATA ANALYSIS

We present an experience-driven method that teaches data agents to use charts within the process of agentic data analysis. The method distills analysis experience and representation experience from trajectories generated under different representation conditions. The validated experiences then guide the agent in selecting analytical operations and evidence representations, allowing charts to support intermediate reasoning rather than serve only as final outputs.

We first describe the structure and contents of the experience package. We then present conditioned trajectory rollouts, trajectory scoring and attribution, and the iterative procedure for experience distillation and validation. Finally, we explain how the resulting experience package guides downstream analysis.

3.1 EXPERIENCE PACKAGE

Storage and tool layers. The experience package consists of a storage layer and a tool layer. The storage layer uses a file-system structure in which an entry document organizes independent experience documents. Each document contains metadata fields such as name, description, and category, together with its experience content, which supports direct reading, editing, and versioning.

The tool layer provides Read, Create, Edit, and Bash for reading, creating, modifying, and managing experience documents. The builder agent uses all four tools to maintain the package, whereas the data agent accesses experience only through Read. Experience is loaded progressively. The data agent first receives document metadata and then reads relevant content according to the current task and analysis state.

Analysis and representation experiences. We extract two types of experience. Analysis experience $\mathcal{E}_{\text{ana}}$ summarizes how to select, organize, and verify analytical operations. Representation experience $\mathcal{E}_{\text{repr}}$ summarizes when to use text, charts, or hybrid representations and how to use them to advance

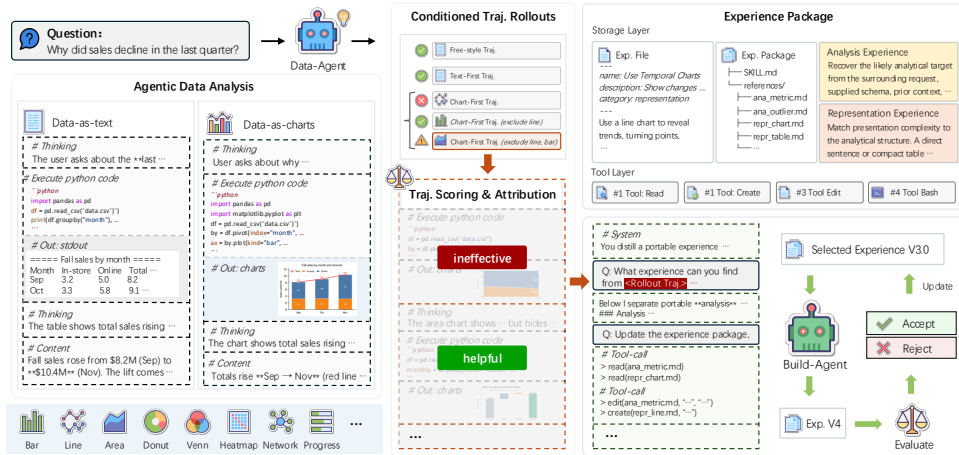


Figure 2: Overview of the experience-driven method for chart use in agentic data analysis. A data agent produces conditioned trajectories with different representation preferences, and a trajectory judge scores their outcomes and attributes intermediate decisions. A builder agent distills analysis and representation experiences into a validated experience package that guides downstream analysis.

subsequent analysis. The two types are stored separately and updated and validated as one experience package. We denote its v -th version by $\mathcal{E}^{(v)} = (\mathcal{E}_{\text{ana}}^{(v)}, \mathcal{E}_{\text{repr}}^{(v)})$.

3.2 CONDITIONED TRAJECTORY ROLLOUTS

We denote the i -th PROBETASK by $p_i = (D_i, q_i, z_i, y_i^*)$, where the four elements represent the data, question, task context, and reference answer. For each task, we generate multiple analysis trajectories under different representation conditions while holding the base agent, tools, and interaction budget fixed. The trajectory generated under condition ρ is denoted by τ_i^ρ . Each task yields a bundle of five trajectories,

$$\mathcal{B}_i = \left\{ \tau_i^{\text{U}}, \tau_i^{\text{T}}, \tau_i^{\text{C}_1}, \tau_i^{\text{C}_2}, \tau_i^{\text{C}_3} \right\}. \quad (1)$$

Here, τ_i^U is an unconstrained rollout, τ_i^T is a text-preferred rollout, and $\tau_i^{C_k}$ for $k \in \{1, 2, 3\}$ are three chart-preferred rollouts. These conditions change only the representation preference and do not restrict the analytical operations available to the agent.

To increase diversity among chart trajectories, we sequentially exclude chart types used by preceding chart-preferred rollouts. The second rollout cannot reuse any chart type from the first, and the third cannot reuse types from either of the first two. This constraint diversifies the visual analyses without assuming that charts outperform text or that any chart type is intrinsically superior. The five trajectories from the same PROBETASK form the basic unit for subsequent scoring and experience distillation.

3.3 TRAJECTORY SCORING AND ATTRIBUTION

For each trajectory, a trajectory judge produces an outcome score and step-level attribution using the task question and reference answer y_i^* . The outcome score evaluates the final answer. The attribution labels key steps as `Helpful`, `Ineffective`, or `Harmful`, and diagnoses the contributions of both analytical operations and representation choices. Together, these signals indicate which behaviors the builder agent should retain or avoid.

3.4 EXPERIENCE DISTILLATION AND VALIDATION

We split the experience-construction and validation sets by PROBETASK, ensuring that all trajectories from the same task remain in the same split. At each iteration, we sample b tasks from the experience-construction set and provide their complete trajectory bundles, outcome scores, and step-level attributions to the builder agent.

Each update has two steps. During insight induction, the builder agent compares trajectories and their attributions to identify transferable analysis and representation patterns. During experience consolidation, it reads the current package and merges these patterns into existing documents or creates new analysis and representation experiences.

The builder agent edits the previously accepted package $\mathcal{E}^{(v-1)}$ and produces a candidate $\tilde{\mathcal{E}}^{(v)}$. Let $Q(\mathcal{E})$ denote validation performance with experience package $\mathcal{E}$. The update rule is

$$\mathcal{E}^{(v)} = \begin{cases} \tilde{\mathcal{E}}^{(v)}, & Q(\tilde{\mathcal{E}}^{(v)}) > Q(\mathcal{E}^{(v-1)}), \\ \mathcal{E}^{(v-1)}, & \text{otherwise.} \end{cases} \quad (2)$$

The candidate is accepted only when validation performance strictly improves. After multiple iterations, the final package $\mathcal{E}^*$ is frozen for downstream tasks.

3.5 EXPERIENCE-GUIDED ANALYSIS

For downstream tasks, the data agent uses the frozen package $\mathcal{E}^*$. The system first provides document metadata, after which the agent reads relevant experiences according to the question and current analysis state. Analysis experience guides the selection and verification of analytical operations. Representation experience guides whether intermediate results use text, charts, or both. The base model parameters remain unchanged. The external experience package guides chart use according to the task and analysis state, making charts part of the analytical process rather than a fixed preference or final presentation artifact. Figure 2 provides an overview of the full pipeline.

4 EXPERIMENTS

We evaluate our method on three benchmarks covering analytical report generation, long-horizon exploration, and exact-answer data analysis. We first describe the benchmarks and evaluation protocol, then report the available results and retain incomplete runs as explicit TBD placeholders.

4.1 BENCHMARKS

We use three benchmarks with complementary task forms and follow the evaluation protocol and metric computation defined by each benchmark. **DAComp-DA** contains 100 open-ended tasks over potentially multi-table SQLite databases and requires reports with analysis, conclusions, and visualizations (Lei et al., 2025); we report Completeness, Accuracy, Insightfulness, Readability, Analytical Depth, Visualization, and Overall. **DDR-Bench** evaluates long-horizon exploration without predefined queries (Liu et al., 2026); we use its public 10-K setting over SEC disclosures from 100 companies and report Message-Wise accuracy, Trajectory-Wise accuracy, and Overall. **DSBench** derives exact-answer tasks from ModelOff competitions with spreadsheet, text, and image inputs (Jing et al., 2024); its Data Analysis subset contains 40 challenges and approximately 466 questions, and we report Task-level Accuracy, Competition-level Accuracy, and Overall.

4.2 EXPERIMENTAL SETUP

We compare reported benchmark agents, a baseline agent reproduced in our common harness, and our method across Claude Opus 5, Claude Sonnet 5, GPT-5.6 Luna, GPT-5.6 Sol, DeepSeek V4.1 Flash, and HY4 Preview. Within each backbone, the baseline and our method share the model version, decoding configuration, system instructions, data access, interaction budget, ReAct loop, Bash, Python, and SQL tools, stdout, stderr, and chart feedback; only our method reads the frozen experience package, so improvements cannot be attributed to plotting access alone.

All experience distillation uses only PROBETASK and follows the conditioned trajectory rollout, trajectory scoring and attribution, and validation procedures in Section 3.4. Source-data flags remain within a single PROBETASK split, and all external benchmarks are held out from experience construction to prevent source leakage and assess transfer. Each backbone constructs its experience package independently from the same initial structure, and the accepted version is frozen before external evaluation.

Table 3: Results on *DAComp-DA*.

Method	LLM Backbone	Completeness	Accuracy	Insightfulness	Readability	Analytical Depth	Visualization	Overall
OpenHands (reported)	GPT-5	60.98	40.3	49.39	35.51	69.80	21.40	46.99
	DeepSeek-V3.1	49.88	33.25	41.66	36.00	33.20	11.00	33.87
DA-Agent (reported)	GPT-5	64.23	43.81	56.89	43.59	76.80	27.44	50.84
	DeepSeek-V3.1	48.74	32.97	42.43	37.25	35.00	11.45	34.33
Baseline (ReAct, Reproduced)	Claude Sonnet 5	70.27	52.55	68.85	90.80	89.00	62.80	69.84
	GPT-5.6 Luna	66.21	46.61	65.45	83.83	80.74	7.66	54.32
	GPT-5.6 Sol	71.74	58.04	72.42	94.60	93.00	61.80	73.24
	DeepSeek V4.1 Flash	83.91	67.80	81.77	98.0	94.44	75.72	82.23
OURS	Claude Sonnet 5	TBD	TBD	TBD	TBD	TBD	TBD	TBD
	GPT-5.6 Luna	68.10	51.18	65.24	82.83	85.00	27.61	59.78
	GPT-5.6 Sol	TBD	TBD	TBD	TBD	TBD	TBD	TBD
	DeepSeek V4.1 Flash	TBD	TBD	TBD	TBD	TBD	TBD	TBD

Table 4: Results on *DDR-Bench*.

Method	LLM	Msg. (%)	Traj. (%)	Overall (%)
ReAct (reported)	Claude Sonnet 4.5	77.61	60.58	69.10
	GPT-5.1	37.12	44.25	40.69
	GPT-5.2	44.89	41.09	42.99
	DeepSeek V3.2	60.08	38.15	49.12
Baseline (ReAct, Reproduced)	Claude Sonnet 5	61.2	52.2	56.7
	GPT-5.6 Luna	44.5	45.1	44.8
	GPT-5.6 Sol	60.2	58.5	59.4
	DeepSeek V4.1 Flash	64.2	59.0	61.6
OURS	Claude Sonnet 5	TBD	TBD	TBD
	GPT-5.6 Luna	TBD	TBD	TBD
	GPT-5.6 Sol	TBD	TBD	TBD
	DeepSeek V4.1 Flash	TBD	TBD	TBD

Table 5: Results on *DSBench*.

Method	LLM	Task (%)	Comp. (%)	Overall (%)
Code Inter. (reported)	GPT-3.5	11.16	8.23	9.70
	GPT-4o	23.82	22.65	23.24
AutoGen (reported)	GPT-3.5	20.82	13.80	17.31
	GPT-4o	34.12	26.72	30.42
Baseline (ReAct, Reproduced)	Claude Sonnet 5	88.47	88.42	88.45
	GPT-5.6 Luna	75.11	73.84	74.47
	GPT-5.6 Sol	89.9	89.4	89.7
	DeepSeek V4.1 Flash	TBD	TBD	TBD
OURS	Claude Sonnet 5	TBD	TBD	TBD
	GPT-5.6 Luna	76.72	76.25	76.49
	GPT-5.6 Sol	TBD	TBD	TBD
	DeepSeek V4.1 Flash	TBD	TBD	TBD

Each benchmark uses its official evaluator, namely *DAComp-DA*’s official rubric and GSB suite, *DDR-Bench*’s GPT-5-mini checker, and *DSBench*’s answer-correctness procedure. Methods share samples, judge prompts, and parameters within each benchmark, and reported prior results are compared only when splits and protocols match. All core comparisons are paired, with clustered bootstrap confidence intervals at the task, company, and challenge levels, respectively.

4.3 MAIN RESULTS

We report each benchmark under its official metric and do not aggregate scores across incompatible scales. Cells marked “TBD” are placeholders for incomplete runs or official evaluations, not estimates.

DAComp-DA. Table 3 reports analytical report generation results. We will report official Overall from [A] to [B], together with changes in Completeness $[\Delta C]$, Accuracy $[\Delta A]$, and Visualization $[\Delta V]$. Joint gains in content-correctness metrics and Visualization would indicate that visual representation supports analysis rather than surface presentation alone.

DDR-Bench. Table 4 reports long-horizon exploration results. We will report Message-Wise accuracy $[X.X]$ and Trajectory-Wise accuracy $[Y.Y]$, with Baseline-relative gains of $[\Delta X]$ and $[\Delta Y]$ points. Larger Trajectory-Wise gains would indicate a stronger effect on long-horizon exploration and evidence synthesis, whereas gains in both metrics would indicate improvements in both local discovery and end-to-end coverage.

DSBench. Table 5 reports exact-answer analysis results. We will report Task-level Accuracy $[X.X]$ and competition-macro Competition-level Accuracy $[Y.Y]$, with Baseline-relative gains of $[\Delta X]$ and $[\Delta Y]$ points. Concurrent gains would indicate that improvements are not driven only by competitions containing more tasks.

4.4 ABLATION STUDIES AND MECHANISM ANALYSIS

We conduct all ablation studies on *DAComp-DA* using the same backbone, ReAct loop, tool interface, interaction budget, and official evaluator. We report Acc., Vis., and Overall, which respectively measure analytical correctness, visual presentation quality, and aggregate performance. Unless

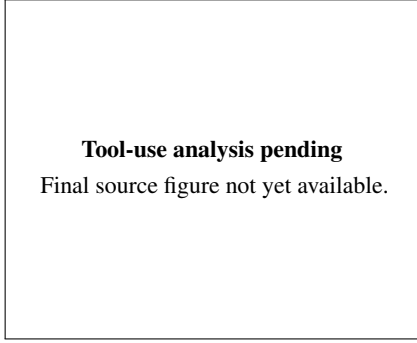


Figure 3: Tool-use on DAComp-DA (TBD).
ReAct, Bash, and experience reads.

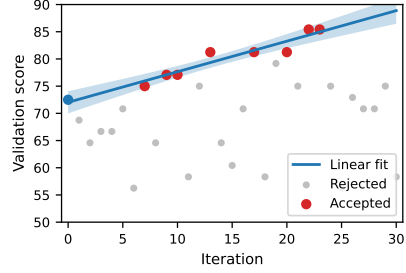


Figure 4: Validation scores during experience-package distillation.

otherwise stated, the ablations change only the component under study. Tables 6–8 report the currently available results.

Experience construction strategies. Table 6 compares the baseline, one-step distillation, and our method. One-step distillation constructs a frozen package from the complete trajectory corpus in one builder call, whereas our method updates the package over successive batches with validation. The current runs cover 100 tasks for one-step distillation and 92 for our method; their similar Acc., higher Overall for one-step distillation, and higher Vis. for our method remain preliminary until evaluation on a matched task set.

Mechanisms of iterative distillation. Table 7 isolates conditioned trajectory evidence and validation-based package selection. The rollout-free variant receives the same questions, data descriptions, and reference insights without trajectories, outcome scores, or step-level attributions, while the ungated variant retains every update; results remain TBD.

Analysis and representation experiences. Table 8 compares the complete package with analysis-only and representation-only variants obtained by retaining documents from only the corresponding category. Acc. measures effects on analytical correctness, while Vis. measures effects on visual evidence and presentation; results remain TBD.

Tool use and optimization dynamics. Figure 3 will compare ReAct rounds, Bash calls, and experience-read calls for the baseline, one-step distillation, and our method, with results currently TBD. Figure 4 reports the available optimization trajectory, where validation filters lower-scoring candidates and maintains a non-decreasing trajectory for the retained experience package.

5 RELATED WORK

5.1 DATA AGENTS.

LLM-driven data agents increasingly follow a plan–execute–observe–revise paradigm. DS-Agent couples language-model reasoning with an executable analysis environment and retrieves historical cases to support data-science problem solving; Data Interpreter organizes complex tasks as dynamic directed acyclic graphs, advancing analysis through hierarchical decomposition and code execution; AIDE models machine-learning solution development as tree search in code space; DS-STAR

Table 6: Ablation to experience construction, comparing baseline and one-step distillation (OSD).

Variant	Acc. (%)	Vis. (%)	Overall (%)
Baseline	46.61	7.66	54.32
OSD	51.26	26.35	61.10
OURS	51.18	27.61	59.78

Table 7: Ablation to conditioned rollouts (CR) and validation gating (VG) in iterative distillation.

Variant	Acc. (%)	Vis. (%)	Overall (%)
w/o CR	TBD	TBD	TBD
w/o VG	TBD	TBD	TBD
OURS	TBD	TBD	TBD

Table 8: Ablation to experience categories, retaining analysis experience only (Ana.) or representation experience only (Rep.).

Variant	Acc. (%)	Vis. (%)	Overall (%)
Ana.	TBD	TBD	TBD
Rep.	TBD	TBD	TBD
OURS	TBD	TBD	TBD

handles heterogeneous data-analysis tasks via iterative planning and plan-sufficiency verification; and AutoKaggle and related systems adopt multi-agent division of labor for data understanding, solution planning, program development, and result review (Guo et al., 2024; Hong et al., 2025; Jiang et al., 2025; Nam et al., 2025; Li et al., 2025). Meanwhile, DeepAnalyze targets end-to-end coverage of data preparation, analysis, modeling, visualization, and report generation through specialized training, and InsightBench extends evaluation to open-ended, multi-step business insight generation (Zhang et al., 2025; Sahu et al., 2025). Although these lines of work continue to advance planning, search, collaboration, and training mechanisms, their analysis loops still rely predominantly on textual representations of structured data and on code-execution outputs as intermediate feedback. Some systems can generate charts, but few isolate whether agents inspect self-generated charts and leverage the visual patterns they reveal to form hypotheses and steer subsequent analysis. We focus on how visual evidence enters the intermediate reasoning loop of data agents.

5.2 CHARTS IN DATA ANALYSIS.

In parallel with data agents, research on charts has made substantial progress in chart generation, understanding, and representation choice. For chart generation, LIDA established a pipeline from data summaries and visualization goals to plotting code; MatPlotAgent uses rendered visual feedback to iteratively repair visualization programs; ChartMimic, AMACE, and MultiVis-Agent further improve reliability through cross-modal reconstruction, multi-agent collaboration, and iterative optimization; and ChartOptimiser shows that chart design should adapt to the analytical task at hand (Dibia, 2023; Yang et al., 2024; 2025; Namgoong et al., 2025; Lu et al., 2026; Wang et al., 2025). For chart understanding, UniChart, ChartLlama, and ChartBench advance chart reading and reasoning via specialized training and large-scale evaluation; DePlot, MatCha, and TinyChart improve numerical analysis through derendering, structural recovery, and programmatic reasoning; and ChartAgent and Charts-of-Thought demonstrate that visual tools combined with structured reasoning can strengthen complex chart analysis (Masry et al., 2023; Han et al., 2023; Xu et al., 2023; Liu et al., 2023a;b; Zhang et al., 2024; Kaur et al., 2026; Das et al., 2025). Building on this foundation, FRES, Format Matters, and TableDART study effectiveness differences and dynamic selection among textual, image, and chart representations, while systems such as EvidFuse construct visual evidence on demand during analysis or reporting (Zhou et al., 2025; Ho et al., 2026; Xing et al., 2026; Lin et al., 2026). However, existing work typically generates user-specified charts, interprets externally provided charts, or selects among pre-existing representations; it has not systematically studied how data agents jointly choose analytical operations and text or chart representations, and use self-generated charts as intermediate evidence for discovering patterns and advancing subsequent analysis.

6 CONCLUSION

We study how charts can serve as internal epistemic actions rather than end-of-analysis illustrations. PROBETASK shows that current data agents can complete end-to-end analysis through self-generated charts, yet often fail to turn plotting-tool access into consistent gains, and exhibit strong model-specific representation preferences when text and charts are both available. Task slices indicate complementary roles: charts support trends and pattern discovery, whereas text better preserves precise, verifiable evidence. Our method distills analysis and representation experiences from conditioned trajectory rollouts with outcome scores and step-level attributions, learning both what evidence to construct and how to represent and use it without fixed representation-selection rules. Experiments on held-out benchmarks indicate that this experience-driven chart use improves over matched baselines, supporting the view that representation choice is part of the analytical policy rather than a cosmetic output.

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A COMPLETE PROBETASK SLICE RESULTS

Tasks may have multiple analytical-operation labels, while each task has one primary cognitive-operation label. Low- n rows are included for completeness and should not be interpreted as stable representation effects.

Table 9: Scores by analytical operation.

Slice	n	No-tool	Text	Chart	Free-style
Filtering	109	0.257	0.752	0.752	0.755
Feature derivation	116	0.257	0.739	0.752	0.744
Aggregation	163	0.305	0.773	0.791	0.781
Normalization	21	0.143	0.571	0.512	0.583
Sorting	22	0.284	0.818	0.830	0.784
Alignment/joining	8	0.188	0.719	0.563	0.719
Temporal change	37	0.237	0.757	0.838	0.757
Distributional stats.	1	1.000	1.000	1.000	1.000
Statistical inference	10	0.200	0.925	0.900	0.900
Forecasting	3	0.000	1.000	1.000	1.000

Table 10: Scores by primary cognitive operation.

Slice	n	No-tool	Text	Chart	Free-style
Lookup	12	0.146	0.604	0.521	0.521
Scan and selection	24	0.219	0.958	0.958	0.958
Comparative judgment	69	0.366	0.801	0.783	0.808
Pattern recognition	54	0.306	0.769	0.833	0.764
Relational explanation	17	0.221	0.706	0.735	0.765
Multidimensional integration	20	0.150	0.575	0.650	0.600
Conditional reasoning	4	0.813	1.000	0.750	1.000

B ITEMS REQUIRED BEFORE SUBMISSION

The final manuscript still requires: confirmation that the method terminology and PROBETASK do not conflict with prior names; final probe and framework artwork; a qualitative trajectory figure; pseudocode for experience distillation and experience-guided analysis; completed external-benchmark and ablation results with statistical tests; judge robustness and human-agreement checks; representative experience entries and full prompts; and compute, cost, and implementation details.